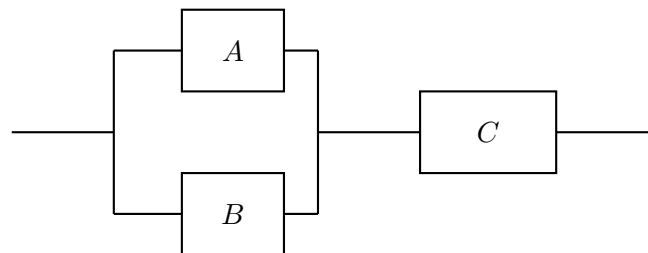


1 Learning task 1: the role of statistics in engineering

We start with two examples from the bachelor Military Systems and Technology.

Maintenance: the reliability of a circuit

A circuit operates only if there is a path of functional devices from left to right. Devices A and B are connected in parallel, device C is in series.



Figuur 1: *

Figure 1.1

Circuit with three devices

For example, the circuit operates if the devices A and C function, but the circuit fails, if device C does not function. Assume that all devices function on average for 10 hours. What is the probability that the circuit will operate for at least 9 hours?

We look at this problem in two different ways, first deterministic (i.e. there is no variability), then stochastic (i.e. there is variability).

- In the deterministic setting, all devices will function for precisely 10 hours. The requested probability is therefore equal to 1.
- In the stochastic case, both a simulation using Excel and a calculation (later in this course) result in a probability of approximately 0.26.

See the Excel-file "PMTPS_STAT_1_stochastic.xlsx" or the print screen in Figure 1.2.

In this file, the above circuit is simulated 10.000 times. In each row, three life times for the devices A , B and C are generated. The life time of the circuit is calculated in column "D". Next, in column "E", it is determined whether the circuit still functions after 9 hours. In that case, the number 1 is shown in column "E".

The requested probability is then calculated in cell "E10003". The magnitude of this probability is partly dependent on the number of simulations. Use the button "F9" to examine the sensitivity at this point.

Operations Research: Search and Detection

An observer is protecting a lane with width D while moving at speed V through the lane according some fixed pattern. Any target that closes the observer to within his sweep radius R is detected. So the observer's detecting device is binary: the target is either detected or not detected.

	A	B	C	D	E
1	Device A	Device B	Device C	Circuit	> 9
9988	6,841	14,873	8,209	8,209	0
9989	4,240	8,650	15,175	8,650	0
9990	18,044	6,365	38,085	18,044	1
9991	1,194	1,557	5,246	1,557	0
9992	38,282	8,686	0,611	0,611	0
9993	7,397	14,077	18,563	14,077	1
9994	4,863	1,553	14,752	4,863	0
9995	6,554	5,390	2,647	2,647	0
9996	6,162	32,267	10,679	10,679	1
9997	3,081	1,041	11,934	3,081	0
9998	15,068	5,047	97,078	15,068	1
9999	3,645	0,658	5,429	3,645	0
10000	9,057	14,896	0,006	0,006	0
10001	3,125	17,618	7,873	7,873	0
10002	Mean A	Mean B	Mean C	Mean Circuit	Probability
10003	9,983	10,122	9,876	6,644	0,2638

Figuur 2: *
Figure 1.2
Life length of a circuit

We assume that targets intend to traverse this lane southwards. We also assume that the target speed U is constant along its path. A further assumption is that we know the intent and the capabilities of the target. More precisely, we assume that we know its speed. Its position, however, is unknown, so arbitrary. In stochastic terms: all positions have an equal probability of occurrence.

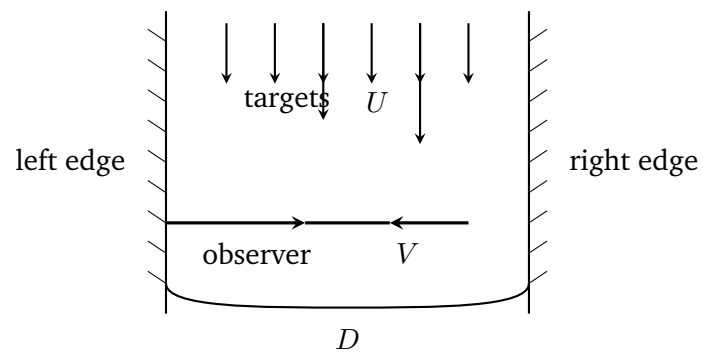
Since the target knows that a preference for the edges (i.e. positions nearby the edges of the lane are more likely than positions in the centre of the lane) leads to a lower probability of detection, the target chooses its position based on the probability function in Figure 1.4.

So if $D = 20$ nm, the probability that the target chooses a position between -5 nm and $+5$ nm is equal to 0.25 (i.e. the area under the graph of the probability function between $x = -5$ and $x = +5$), while the probability that the target chooses a position between -10 nm and -5 nm or between 5 nm and 10 nm is equal to 0.75.

What is the probability that the target will be detected?

Using probability, calculus and geometry, it is not difficult to derive the following formula

$$P_{\text{det}} = \left\{ \frac{4R^3}{3D^3} \cdot \left(1 + \left(\frac{V}{U} \right)^2 \right) - \frac{2R^2}{D^2} \cdot \sqrt{1 + \left(\frac{V}{U} \right)^2} + \frac{2R}{D} \right\} \cdot \sqrt{1 + \left(\frac{V}{U} \right)^2}.$$



Figuur 3: *

Figure 1.3

Linear barrier (reversing course at the edge)

Example 1

$$D = 20$$

$$R = 4$$

$$V = 18$$

$$U = 12$$

$$P_{\text{det}} = 0.5236$$